

KHALEELUR RAHMAN

AI Developer

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📍 Central, Chennai

PROFESSIONAL SUMMARY

Passionate AI & Machine Learning Engineer with 1+ years of professional experience in building and deploying real-world ML/DL models across domains such as face recognition, NLP chatbots, and video analytics. Skilled in developing scalable AI systems using FastAPI, Rasa, Hugging Face, and OpenCV. Continuously learning and innovating to deliver efficient, data-driven solutions.

TECHNICAL SKILLS

- **Frontend:** React.js, Electron.js, Tailwind CSS
- **Backend:** Python, FastAPI, REST APIs, RAG, Agentic AI, QLoRA
- **AI / ML:** Computer Vision, Face Recognition, YOLO, NLP, Speech-to-Text
- **Databases:** PostgreSQL, MongoDB, MSSQL
- **Deployment & DevOps:** Docker, Linux Server Deployment, Nohup Python Deployment, RTSP Handling
- **Backup & Version Control:** Git, GitHub, Server Backup & Recovery

WORK EXPERIENCE

Video Management System

Jan 2025- Present

- Designed and developed an AI-powered Video Management System with **23 real-time intelligent detections**.
- Implemented **face recognition, intrusion & perimeter detection, crowd analysis, object & vehicle detection, and women & infant safety alerts**.
- Built real-time video streaming and AI event monitoring for multiple cameras.
- Developed a cross-platform desktop application for live monitoring and camera management.
- Supported dynamic camera onboarding, scalability, and production deployment.

Tech Stack: Python, FastAPI, OpenCV, Deep Learning, React.js, Electron, MongoDB

face recognition system

May 2025- July 2025

- Designed and developed a full-scale face recognition and video intelligence platform from scratch using **Dlib and InsightFace**.
- Implemented real-time face detection, tracking, and recognition with a Face Gallery supporting **registration, bulk enrollment, and image augmentation**.
- Built **face event and investigation** workflows, including match/no-match detection and watchlist-based alerts across historical video data.
- Integrated multi-camera RTSP video processing with efficient stream management for concurrent feeds.
- Developed secure **role-based access control (RBAC)**, live camera dashboards, and real-time alerting systems.
- Built **FastAPI-based REST services**, optimized for low-latency, high-throughput processing, and deployed the **Dockerized** system to production servers.

Tech Stack: Python, FastAPI, OpenCV, Dlib, InsightFace, PostgreSQL/MongoDB, Docker

- Developed an AI-powered chatbot for pension FAQs, status queries, and form guidance
- Implemented **RAG with Vector Database** for document-based question answering
- Fine-tuned NLP models using **QLoRA & Hugging Face Transformers**
- Built FastAPI backend integrated with MSSQL
- Reduced manual support workload through automation
- Handled production deployment and server maintenance

Tech Stack: Python, FastAPI, QLoRA, Hugging Face, RAG, Vector DB, MSSQL

BOCW-AI Chatbot

Oct 2025- Nov 2025

- Designed and developed an AI chatbot for Bihar BOCW (Building & Other Construction Workers Welfare Board).
- Automated welfare **scheme inquiries, eligibility checks, and beneficiary guidance.**
- Implemented secure backend APIs for government data handling.
- Ensured chatbot scalability for public-facing citizen usage.
- Independently handled development, testing, and server deployment.

Tech Stack: Python, NLP, FastAPI

AI face attendance system

Oct 2025- Nov 2025

- Designed and developed a complete AI-based face attendance system from scratch.
- Implemented real-time face detection & recognition with live.
- Enabled **high-accuracy recognition without model retraining using image augmentation** during user registration.
- Built **bulk face registration** pipelines for large employee datasets.
- Developed REST APIs for **user registration, attendance tracking, and reporting.**
- Designed audit-ready attendance databases and handled Dockerized production deployment.

Tech Stack: Python, FastAPI, OpenCV, Dlib, InsightFace, PostgreSQL

TNCSC-AI -Face attendance & ASR

Dec 2025- Jan 2025

- Developed an AI-based face attendance system for TNCSC.
- Implemented speech-to-text processing for government meetings.
- Built a meeting transcription system with: **Speaker identification , Speaker-wise transcript storage (Speaker 1, Speaker 2, etc.)**
- Designed secure storage for **voice data and transcriptions.**
- Ensured compliance with government portal and data security standards.
- Fully deployed and maintained the system on production servers.

Tech Stack: Python, Speech-to-Text, NLP, Deep Learning, FastAPI

EDUCATION

Anna University | 2020-2024

Rajalakshmi Institute of Technology

BTech -Artificial Intelligence and Data Science

LANGUAGE

English,Tamil